

Darius Mahjoob

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EDUCATION

University of Southern California, Viterbi School of Engineering

GPA: 3.9/4.0

Master of Science in Electrical Engineering and Machine Learning

May 2027

Bachelor of Science in Computer Science and Business, Minor in AI Applications

Honors: Presidential Merit Scholarship, Dean's List (5x), Academic Achievement Award (4x), Thematic Option Honors

Relevant Coursework: Software Engineering, Data Structures, Algorithms, Natural Language Processing, Computer Systems

SKILLS AND INTERESTS

Languages: Python, Java, C/C++, Linux, x86 Assembly, Ruby, SQL, JavaScript, HTML/CSS, SQL

Dev Tools: Git/GitHub, Docker, Postman, Kubernetes, Jest, AWS (EC2/S3/Lambda/Quicksight), Power BI, Tableau, Supabase

AI/ML Libraries: PyTorch, TensorFlow, Keras, NLTK, SpaCy, Cupy, pandas, Matplotlib, NumPy, Numba, scikit-learn, JAX, FLAX

Frameworks: FastAPI, LangChain, Pydantic, Apache Tomcat, Websockets, RESTful APIs, React, Node.js, Linear, Agile & Scrum

EXPERIENCE

Hemut (YCx25)

Los Angeles, CA

Technical Project Manager & Software Engineer

June 2025-Present

- Developing an AI-powered TMS to automate routing and load matching using voice agents, improving scheduling time by 90%
- Integrating predictive sequential machine learning models for fleet optimization, increasing route selection efficiency by 42%.

DrFirst, Inc.

Remote, USA

Software Engineering & AI Intern

June 2026-August 2026

- Built an [AI pipeline](#) to analyze 430+ marketing assets using Meta's TRIBEv2 brain-encoding model to predict neural responses.
- Mapped 20K+ neural activation values per input into emotion classifications, enabling the marketing team to evaluate and refine creative assets before launching campaigns.

University of Southern California, Viterbi School of Engineering

Los Angeles, CA

Teaching Assistant: Neural Networks, Data Structures & Algorithms, Machine Learning, Data Mining

December 2024-Present

- Revise [syllabi](#) and [assignments](#) to cover AI models in Keras/TensorFlow, data preprocessing/fine-tuning/evaluation in pandas.
- Built a real-time Q&A system with FastAPI & WebSockets to instantly route student questions to professors, deployed on Render.

USC Interactive Collaborative & Autonomous Robotics Lab (ICAROS)

Los Angeles, CA

Undergraduate Student Researcher - Machine Learning

November 2024-Present

- Built a Variational Autoencoder with JAX/FLAX to train reinforcement learning agents on synthetic (LLM-generated) data.
- Preprocessed and cleaned data using transformers (Sentence-BERT) to categorize inputs, optimizing model accuracy by 36%.

DrFirst, Inc.

Remote, USA

Data Engineering & Machine Learning Intern

June 2025-August 2025

- Automated an ETL pipeline for 53 pharmaceutical brands, integrating AWS S3, Lambda, and QuickSight to visualize prescription analysis data and reduce manual processing time by more than 90%.
- Implemented an LSTM neural network to model and forecast brand-level data growth based on historical data.

Captis Intelligence, Inc.

Remote, USA

Software Engineer Intern

June 2024-August 2024

- Wrote a Python program to increase a chatbot's ability to comprehend user queries by 25% within the Captis database.
- Transformed a pre-trained LLM to generate efficient SQL queries from a dataset of over 5 million prompts.

PROJECTS

Pokémon [Desolation](#), [Azure](#), [Rejuvenation](#), & [Reborn](#) | C++, Ruby, Python, GitHub, RPGMakerXP

Head Developer & Lead Composer

- Design and produce several functional unofficial Pokémon games, incorporating 5000+ custom sprites, 300+ maps, 1600+ events, enemy AI algorithms, and original gameplay mechanics to enhance player experience.
- Reach over 1 million downloads and composed 100+ original musical tracks implemented throughout the games.

[Clarify - A Live Classroom Q&A System](#) | Python FastAPI, Agentic AI, RAG, WebSockets, Supabase

- Deployed a real-time Q&A platform enabling live student/professor interaction, text-to-speech input, and AI-generated responses.
- Implemented an agentic AI pipeline leveraging LLMs, prompt engineering, and Retrieval-Augmented Generation with SentenceTransformers to summarize questions, categorize topics, flag urgency, and generate quizzes from course material.

[Meeting Transcription Discord Bot](#) | Python, Pycord, Google Speech-to-Text API, Groq API, FFmpeg

- Developed a Discord bot to automatically record, transcribe, and summarize virtual meetings.
- Introduced customizable input dictionaries, enabling users to tailor AI summarization to project-specific terminology.

[ML Stock Prediction Model](#) | Python Flask, ARIMA Neural Networks, TensorFlow, Keras, REST APIs

- Designed a deep learning model using Python and TensorFlow to predict stock trends, increasing prediction accuracy to 71%.
- Built a Flask web interface for real-time stock visualization, ensuring seamless user interaction and data accessibility.